R&D Pathways and Innovation

How R&D Works in This Simulation

Research and development is the primary mechanism by which firms improve their products, reduce manufacturing costs, and build long-term competitive advantage. R&D spending each quarter contributes to R&D capital -- an intangible asset that accumulates over time and determines when (and whether) a firm achieves the next technology generation.

The R&D Pipeline

Three Parallel Research Programs

Each firm can invest in three distinct R&D programs simultaneously. Resources (dollars) are allocated across these programs each quarter:

1. Product R&D (Efficacy & Safety)

Goal: Develop the next-generation SRT compound with better efficacy and fewer side effects.

What it produces:

• Progress toward Gen 2, Gen 3, and Gen 4 compounds.

• Each generation requires crossing a cumulative R&D investment threshold (see below).

• Progress is NOT guaranteed -- there is a stochastic element. More spending increases
the probability of success but does not guarantee it.

Key milestones:

Transition	Cumulative Investment Req	Success Probability	Expected Timeline
Gen 1 -> Gen 2	$400M–$600M	80% once threshold reache	12–16 quarters
Gen 2 -> Gen 3	$800M–$1.2B	65% once threshold reache	12–20 quarters
Gen 3 -> Gen 4	$1.5B–$2.5B	50% once threshold reache	16–24 quarters

How the stochastic element works: Each quarter, if cumulative product R&D spending exceeds the minimum threshold for the next generation, the environment draws a success check:

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P(success this quarter) = base_rate * (cumulative_spend / threshold)^0.5
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Where base_rate is ~10–15% per quarter once the minimum is met. Spending well beyond the threshold increases the probability but with diminishing returns.

If the check fails, spending is not lost -- it continues to count toward future checks. But it means a firm could spend $600M and still not achieve Gen 2 for several additional quarters, while a luckier competitor might succeed at $450M.

When a generation advance occurs:

• The firm's product quality scores (efficacy, safety) jump to the new generation's
parameters.

• The firm must retool its manufacturing process (1–2 quarters of reduced capacity
during changeover).

• The firm can market the new generation immediately.

• Competitors can observe the advance (it becomes public knowledge).

2. Process R&D (Cost Reduction)

Goal: Reduce manufacturing cost (COGS) within the current generation through process improvements.

What it produces:

• Incremental COGS reduction: each $50M invested reduces COGS by ~3–5% (diminishing returns beyond $200M cumulative per generation).

• Improvements include: better synthesis yields, lower batch failure rates, faster cycle times, raw material substitution.

Key parameters:

Investment Level (cumulative per g	COGS Reduction	Notes
$0–$50M	0–5%	Low-hanging fruit
$50M–$100M	5–12%	Meaningful optimization
$100M–$200M	12–18%	Approaching diminishing returns
$200M+	18–22% max	Hard floor -- need next gen for mo

Process R&D does NOT help achieve the next generation -- it only reduces cost within the current one. A firm must invest in Product R&D to advance generations.

3. Delivery R&D (Convenience)

Goal: Improve the administration route and patient experience.

What it produces:

• Progress toward subcutaneous injection (Gen 2 delivery), oral formulation (Gen 3), or one-time treatment (Gen 4).

• Delivery advances can be achieved independently of product compound advances, but they are harder to achieve without the corresponding compound chemistry.

Key milestones:

Transition	R&D Investment	Difficulty	Standalone Possible?
IV -> Subcutaneous	$100M–$200M	Moderate	Yes (with Gen 1 compound)
Subcutaneous -> Oral	$300M–$500M	High	Requires at least Gen 2 c
Oral -> One-time gene the	$800M–$1.5B	Very High	Requires Gen 3+ compound

Strategic note: A firm could achieve subcutaneous delivery for its Gen 1 compound before a competitor achieves Gen 2. This gives a convenience advantage even with inferior efficacy/safety -- and can be very valuable in the early market where clinic visits are a major barrier.

R&D Spending: Practical Guidance

Quarterly R&D Budget Ranges

Strategy	Quarterly R&D Spend	Annual	Characteristic
Minimal	$5–10M	$20–40M	Mandatory Phase III only;
Conservative	$15–25M	$60–100M	Slow but steady progress
Moderate	$30–50M	$120–200M	Competitive timeline for
Aggressive	$60–100M	$240–400M	Fast-track Gen 2; acceler

Moonshot	$100M+	$400M+	Racing to leapfrog; very

Remember: $10M/quarter is the mandatory Phase III cost (non-discretionary). Any R&D spending above that is the firm's strategic choice.

Allocation Across Programs

There is no "right" split, but common strategies include:

Strategy	Product R&D	Process R&D	Delivery R&D
Balanced	50%	25%	25%
Efficacy-first	70%	20%	10%
Cost leader	30%	55%	15%
Convenience play	30%	15%	55%

Firms must specify their R&D allocation each quarter. The environment tracks cumulative investment in each program separately.

R&D Spillovers and Knowledge

Public Knowledge

When any firm achieves a generation advance, certain knowledge becomes public:

• The fact that the advance is possible (eliminates scientific uncertainty for others).

• General approach (not specific formulation).

• This reduces the remaining R&D cost for competitors by ~15–20% (they can learn
from published literature and reverse-engineer the general approach).

No Explicit Technology Sharing

Firms do not license or share technology in this simulation. Each must develop its own formulation and process. (This could be added as a future extension.)

Academic Breakthroughs (Environment Shocks)

The environment may occasionally generate an "academic breakthrough" shock:

• An open-access scientific publication that advances understanding.

• Reduces all firms' R&D cost for the next generation by 10–20%.

• Probability: ~5% per quarter.

• This is a windfall that rewards firms already investing in R&D (they are closer
to the threshold) and doesn't help firms that have stopped investing.

R&D Risks

Failure Modes

• Clinical failure: A Gen 2 candidate turns out to have unexpected toxicity in
expanded testing. The firm must abandon this compound and restart with a new one.

  • Probability: embedded in the success check (the 20–50% failure probability).

  - Cost: cumulative investment in that failed candidate is partially lost (50% of spending contributes to the next attempt, 50% is sunk).
- Manufacturing translation failure: The new compound works in the lab but cannot be manufactured at commercial scale at acceptable cost.
  - Probability: ~10% per generation advance.
  - Cost: 1–2 quarter delay + $50–100M additional process R&D.
- Regulatory rejection: The FDA doesn't approve the new generation based on the data submitted.
  - Probability: ~5–10% per generation advance.
  - Cost: 2–4 quarter delay while additional data is generated.

The "R&D Trap"

A firm can fall into an R&D trap by:

- Spending aggressively on R&D while competitors also spend aggressively -> no lasting advantage, just mutual cash burn.
- Spending so much on R&D that it cannot fund marketing and operations -> great product with no patients.
- Pursuing Gen 3 before Gen 2 is commercially established -> no revenue base to fund the continued R&D investment.

The optimal R&D strategy depends on competitors' strategies, current market conditions, and the firm's financial position.

Technology Evolution: What Firms Are Racing Toward

The "Golden Product" (Gen 3–4)

The ultimate competitive position is a product that is:

- Highly effective (15+ years of age reversal)
- Very safe (<0.5% serious AE rate)
- Convenient (oral or one-time)
- Cheap to produce (COGS < $2,000/course)

This product can be priced at $5,000–$15,000 per year and serve hundreds of millions of patients. The firm that gets there first (or close to first) with adequate manufacturing capacity captures a market worth hundreds of billions of dollars annually.

The Race Dynamics

- First-mover advantage in R&D is real but not decisive. Being 2–4 quarters ahead in R&D matters, but a competitor can catch up (especially with public knowledge spillovers).
- Speed vs. capital efficiency: Spending $100M/quarter on R&D gets you there faster but requires massive financing. A firm spending $30M/quarter gets there eventually at lower risk.
- The financial market rewards R&D leadership -- equity prices reflect expected future cash flows, and a firm closer to Gen 2/3 commands a premium.

What Firms Should Remember

• R&D is not optional. A firm that does not invest beyond mandatory Phase III will be selling Gen 1 products while competitors sell Gen 2 and Gen 3. Its margin will shrink and its market share will evaporate.

• R&D is not revenue. Dollars spent on R&D do not generate revenue this quarter. The payoff is 3–5 years away. You must finance the gap with operating income, equity, or debt.

• Allocate deliberately. Don't spread R&D evenly across all programs without thought. Choose a focus -- efficacy, cost, or convenience -- based on your strategy.

• Watch competitors' R&D. If a competitor achieves Gen 2 and you haven't, you will lose market share rapidly. The financial-market agent will also devalue your equity.

• R&D has diminishing returns within a generation. Once you've reduced Gen 1 COGS by 20%, further spending on process R&D has minimal effect. Shift resources to the next generation.